

$$\begin{array}{c} ? \\ ? \end{array} \rightarrow \rightarrow \rightarrow \rightarrow \rightarrow \rightarrow \rightarrow \rightarrow Joint-ALNS \rightarrow .$$

$$\begin{array}{c} \mathcal{R}^T_r \\ \mathcal{R}^T_r \\ \mathcal{O}01 \end{array} \rightarrow i_{r1} \rightarrow i_{r2} \rightarrow \cdots \rightarrow i_{rK_r} \rightarrow \mathcal{O}01.$$

$$\begin{array}{c} ? \\ t \end{array} \mathbf{p}_r(t) = (\lambda_r(t), \varphi_r(t), H_r(t)) ,$$

$$(1) \begin{array}{c} \lambda_r(t) \varphi_r(t) \\ H_r(t) \end{array}$$

$$(2) \begin{array}{c} \Delta t = 30\text{s} \\ a,b \\ \mathbf{p}_a = (\lambda_a, \varphi_a, H_a), \mathbf{p}_b = (\lambda_b, \varphi_b, H_b). \end{array}$$

$$(3) \begin{array}{c} \chi_{ab} = \{1, a, b, 0, . \\ ? \\ f \\ d_{ab}^{\text{FS}} = 32.44 + 20 \log_{10} f + 20 \log_{10} d_{ab}. \end{array}$$

$$(4) \begin{array}{c} L^{\text{obs}} \\ L_{ab} = L_{ab}^{\text{FS}} + \chi_{ab} L^{\text{obs}}. \end{array}$$

$$(5) \begin{array}{c} P^{\text{sens}} \\ M^{\text{fade}} \\ P^{\text{req}} = P^{\text{sens}} + M^{\text{fade}}. \end{array}$$

$$(6) \begin{array}{c} a \rightarrow b \\ L_{a \rightarrow b}^{\max} = P_a^t + G_a^t + G_b^r - L^{\text{sys}} - P^{\text{req}}, \end{array}$$

$$(7) \begin{array}{c} P_a^t \\ G_a^t G_b^r \\ L_{\text{sys}}^{\text{sys}} \\ a,b \\ L_{ab}^{\max} = \min \{ L_{a \rightarrow b}^{\max}, L_{b \rightarrow a}^{\max} \}. \end{array}$$

$$(8)$$

$$(9) \begin{array}{c} A_{ab}(t) = \mathbf{1} \{ L_{ab}(t) \leq L_{ab}^{\max} \}. \\ r \\ A_{rG}(t) \\ q \\ A_{rq}(t) \\ q \\ A_{qG}(t) \\ z_{rq}(t) = \\ \frac{1}{t} \\ q \\ r \end{array}$$

$$(10) \begin{array}{c} C_r(t) = A_{rG}(t) \vee \left[\bigvee_{q \in \mathcal{R}^R} z_{rq}(t) A_{rq}(t) A_{qG}(t) \right]. \end{array}$$

$$(11) \begin{array}{c} C_r(t) = 1, r \in \mathcal{R}^T, \\ \sum_{q \in \mathcal{R}^R} z_{rq}(t) \leq 1. \end{array}$$

$$(12) \begin{array}{c} A_{rG}(t) = \\ 0 \end{array}$$

$$c = (\lambda_c, \varphi_c, H_c),$$

$$(13) \begin{array}{c} 0 < H_c - H_c^{\text{DEM}} \leq H_R^{\max}. \\ H^{\text{AGL}} \in \{150, 225, 300\} \text{m}. \end{array}$$

$$\begin{array}{c} q \\ O01 \end{array} \rightarrow c_q \rightarrow \rightarrow O01.$$

$$\begin{array}{c} t_R^{\text{pre}} \\ t^{\text{out}}_q \\ t^{\text{link}}_q \end{array}$$